

Simran Bilkhu

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Oakville, Ontario, Canada

EDUCATION

• McMaster University

Master's in Statistics

September 2024 - April 2026

Hamilton, Canada

◦ Grade: 11.86/12

◦ Thesis [\[link\]](#)

A Comparative Study of Bayesian Factor Models under PLT and UGLT Identification Constraints

• University of Toronto

Specialist in Applied Statistics with a minor in Math (Honours Bachelor of Science)

April 2024

Toronto, Canada

◦ Cumulative GPA (final 2 yrs - 10.5 out of 20.5 total credits): 3.76/4.00

PUBLICATIONS

• Counter-Example to Continuity of Measure in Uncountable Unions

[Link to Preprint](#)

Submitted for review

PROJECTS & PRESENTATIONS

• Statistical Society of Canada Annual Meeting

June 2, 2026

Student Research Poster Presentation Competition (ongoing)

[Link](#)

◦ Scheduled to present thesis research comparing Bayesian factor models under PLT and UGLT identification constraints; submitted as part of the SSC student research competition.

• Bird Migration Count Forecasting

December 2024

Assignment

[Link](#)

◦ Forecasted population trends of *Peregrine Falcons* and *Greater Roadrunners* using satellite climate data: modelled counts (13,642 obs, 2001–2005) from time, location, and weather features.

◦ A spatio-temporal mixed model was used to reduce test error from 1,377 to 31 vs. ridge-penalised Poisson regression, with warmer temperatures positively associated with counts.

• Probability and Stochastics

July 2025

Presentation

[Link to Project](#)

◦ Prepared lectures on martingales, and their relationships to Markov processes, Wiener processes, and Poisson processes.

◦ Provided proofs of several theorems and properties.

◦ Included an introduction to stochastic kernels, and their connection to stochastic matrices for discrete state spaces.

• Bayesian Nonparametrics

January 2025 - April 2025

Assignment

[Link to Project](#)

◦ Provided a thorough proof showing that the proportions from the Chinese Restaurant Process converge in distribution to the GEM distribution.

• Bayesian Nonparametrics

January 2025 - April 2025

Assignment

[Link to Project](#)

◦ Showed that the size-biased permutations of the Poisson-Dirichlet distribution is given by the 2-parameter GEM distribution.

• Statistical Learning

January 2025 - April 2025

Impacts of Diet Intervention on the Gut Microbiome

[Link to Project](#)

◦ Applied dimension reduction via Principal Component Analysis (PCA) on microbiome count data with over 1400 features, and identified relevant taxa.

◦ Used model-based clustering techniques to explore group differences in microbial composition between plant and animal based diets to identify differing microbiome profiles.

- Identified important microbial taxa for dietary classification by analyzing loadings from principal components.

• Research Group Speaker

April 2025

A Review of Sparse Bayesian Factor Analysis when the Number of Factors is Unknown

[Link to Project](#)

- Reviewed and presented work by [Frühwirth-Schnatter et al.](#), highlighting a Bayesian approach to factor analysis with an unknown number of latent factors using spike-and-slab priors.
- Provided a background on shrinkage priors, UGLT structures, and [reversible jump MCMC](#).

• Research Group Speaker

January 2025

A Review of Sparse Principal Component Analysis

[Link to Project](#)

- Presented key theoretical links between factor analysis and PCA, highlighting the benefits of sparse representations for interpretability in high-dimensional settings.
- Compared sparse PCA via penalized optimization (see [Zou et al.](#)) to varimax rotation, using simulation studies to assess interpretability and mean squared error across varying sparsity levels.

• Data Science

November 2024

Establishing Baseline Climate Partitions via Clustering

[Link to Project](#)

- Analyzed time-series raster climate data (temperature, precipitation, elevation) across Canada using python, focusing on spatial patterns and long-term climatic trends.
- Performed K-means and Agglomerative clustering methods to group geographic regions using silhouette analysis to choose the number of clusters.
- Visualized and interpreted clustering results through geographic heatmaps, and cluster-wise summaries.

• Data Science

September 2024

Expansion of Canadian Agriculture and the Relationship with Wetlands in Boreal Shield Provinces

[Link to Project](#)

- Conducted geospatial analysis of 66 GeoTIFF satellite land cover maps (2013–2023) using R's terra package to quantify agricultural land use and wetland area.
- Preprocessed raster data into a tidy format and scaled estimates to square kilometers for year-by-year provincial comparison.
- Designed and deployed a Shiny web app for interactive visualization, allowing users to explore province-specific trends and relationships across time ([link to site](#)).
- Addressed limitations in remote sensing data such as non-random missingness due to satellite imaging challenges, and discussed implications for ecological monitoring and policy.

• Research Group Speaker

September 2024

Hamiltonian Monte Carlo

[Link to Project](#)

- Presented a conceptual and mathematical overview of Hamiltonian Monte Carlo (HMC), explaining the role of Hamiltonian dynamics.
- Co-developed a Shiny web app allowing users to simulate from normal and beta distributions using HMC

WORK EXPERIENCE

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|--|--------------------------------|
| • McMaster University , Research Assistant II | May 2026 - Present |
| • McMaster University , Intermediate Probability Theory (TA) | September 2025 - December 2025 |
| • McMaster University , Statistical Methods and Applications (TA) | January 2025 - April 2025 |
| • McMaster University , Linear Regression Models Using SAS (TA) | September 2024 - December 2024 |
| • Sodexo Canada , IT Intern | Mar 2020 - Aug 2022 |

WORKS CITED

- Frühwirth-Schnatter, Sylvia, et al. "Sparse Bayesian Factor Analysis when the number of factors is unknown (with Discussion)". *Bayesian Analysis*, vol. 20, no. 1, 2025, pp. 213–344.
- Zou, Hui, et al. "Sparse principal component analysis". *Journal of computational and graphical statistics*, vol. 15, no. 2, 2006, pp. 265–86.